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Thy Neighbor's Vote: Peer Effects in Proxy Voting

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Abstract. Institutional investors' proxy voting decisions are influenced by their neighbors. I identify peer effects in proxy voting using close-call votes on shareholder-sponsored governance proposals at publicly traded financial institutions (focal institutions). I first show that the passage of a governance proposal at a focal institution makes the institution more likely to vote against management at its portfolio firms. Using a triple-difference approach, I find that the neighboring institutions of the focal institution that passes a governance proposal become more likely to vote against management in stocks that are heavily held by the focal institution. These results suggest that peer influence is an important determinant of proxy voting behavior.

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Keywords: peer effects • proxy voting • institutional investors • geography • corporate governance

1. Introduction

Proxy voting decisions are characterized by strong strategic complementarities. For a shareholder who prefers a particular vote outcome, the utility from voting in the direction of the preferred outcome increases with the extent that other shareholders vote in the same direction because the likelihood that the vote goes in the desired direction depends on the fraction of shareholders who vote in that direction. Such complementarities create an incentive for shareholders to condition their votes on those of their peers to make their aggregate votes more pivotal. By coordinating their voting behavior, individual institutions can increase the likelihood that the outcome of the vote is in their favor and hence, attain a higher utility from their votes. Anecdotal accounts suggest that institutional investors coordinate with their peers when engaging with companies.¹ Yet, despite its importance in corporate governance, peer effects in shareholder monitoring remain underexplored.

In this paper, I examine peer influence among geographically proximate institutional investors in the context of proxy voting. Influencing peers to vote in a certain way involves a trade-off. The benefit of an increase in the likelihood of a desired vote outcome must be weighed against the costs associated with influencing peers, such as costs incurred to identify like-minded peers, exchange information, and reach agreement about coordinated actions. Although modern communication technologies enable institutions to readily collect hard information about peer institutions from a distance, it has not become easier for institutions to collect “soft” information about their peers,

which by definition, cannot be credibly transmitted (e.g., Stein 2002). Geographic proximity can facilitate the collection of soft information through repeated interactions and face-to-face meetings,² thereby reducing the costs associated with peer influence. Therefore, institutions are likely to derive a higher expected utility from influencing the voting behavior of peers who are geographically more proximate.

Identifying causal peer effects is notoriously difficult. Correlated votes by institutions in a locality may arise because of three reasons using the terminology of Manski (1993). The first is endogenous effects (i.e., peer effects), in which the propensity of an institution to vote in a certain manner is influenced by its neighboring institutions. The second is exogenous (contextual) effects, in which the voting behavior of an institution is correlated with the exogenous characteristics of the environment in which the institution operates (e.g., the economic, political, legal, and social institutions of the locality where the institution is located). The third is correlated effects, in which institutions that are in the same locality behave similarly in their voting decisions because they have similar governance preferences or correlated information. One way to disentangle peer effects from the other two effects is to exogenously change an institution's voting behavior and study its effects on the voting behavior of neighboring institutions.

In this paper, I exploit variation in the voting behavior of publicly traded financial institutions because of close votes on shareholder-sponsored governance proposals at these institutions (hereafter referred to as focal institutions). The passage of a governance proposal at a

focal institution could induce changes in the institution's voting behavior for at least three reasons. First, the corporate governance team at an institution may infer the governance preferences of its shareholders from the outcome of their votes and align its voting decisions with the preferences of its shareholders. Thus, when a majority of an institution's shareholders vote for a governance proposal opposed by management, the institution may take a more proshareholder stance in its own voting decisions. Relatedly, the corporate governance team may be naturally more proactivist and feel emboldened when activist shareholders are more powerful in their own institution. Second, the passage of a shareholder-sponsored governance proposal at an institution signals shareholder dissatisfaction with its management and corporate governance practices, which could prompt the institution to become more active on governance issues at its portfolio firms. Third, there could be a preach what you practice effect, in which institutions that adopt a governance practice cast votes in a manner consistent with that practice. Because the passage of a proposal on a governance policy at an institution increases the likelihood that the institution adopts the policy, the institution may become more likely to support proposals on similar governance policies at its portfolio companies. These considerations suggest that the passage of governance proposals at a focal institution can make the institution more active in voting against management on governance issues at its portfolio companies.

Because the result of a close-call vote around the majority threshold at a focal institution is unlikely to be correlated with contextual or correlated effects faced by the institution, this setting enables me to isolate peer effects from the effects of common unobserved factors. If institutions influence their neighbors in voting, the shock to the voting behavior of a focal institution should be transmitted to its neighbors.

To illustrate, consider two publicly traded financial institutions. Institution A, headquartered in Atlanta, narrowly passes a shareholder-sponsored governance proposal opposed by management, and institution B, headquartered in Boston, narrowly rejects one. The own effect predicts that institution A should become more active in opposing management at its portfolio companies than institution B after the close-vote event than before. If the voting behavior of these two focal institutions influences that of their neighboring institutions, institution A's neighboring institutions (i.e., other Atlanta-based institutions), relative to institution B's neighbors (i.e., other Boston-based institutions), should become more likely to vote against management at their portfolio companies when the companies are exposed to the influence of the focal institutions.

Using a sample of shareholder-sponsored governance proposals at publicly traded financial institutions (i.e.,

focal institutions) that pass or fail within $\pm 2.5\%$ around the majority threshold, I first examine the effect of close-vote events on the voting behavior of focal institutions (the *own* effect). Using a difference-in-differences specification, I find that, relative to focal institutions that narrowly reject a shareholder-sponsored governance proposal, those that narrowly pass such a proposal become more likely to oppose management in their proxy votes. This result is obtained after controlling for close-vote event fixed effects, proposal fixed effects, institution $\times$ proposal type fixed effects, institution $\times$ industry fixed effects, institution $\times$ headquarters state fixed effects, and institution $\times$ firm fixed effects, thereby eliminating the possibility that the results are explained by omitted variables at various levels. The economic magnitude of the own effect is nontrivial. For example, focal institutions that narrowly pass a shareholder-sponsored governance proposal become 2.3–2.7 percentage points more likely to oppose management after the close-vote event than before the event relative to focal institutions that narrowly reject such a proposal. These results suggest that the passage of a governance proposal at a focal institution makes the institution more active in voting against management at its portfolio companies.

If neighboring institutions of a focal institution condition their voting decisions on those of the focal institution, the change in the voting behavior of the focal institution induced by a close-vote event should lead to changes in the voting behavior of its neighbors. I thus examine the effect of a close-vote event at a focal institution on the voting behavior of its neighbors (the *peer* effect). I define neighboring institutions of a focal institution as those headquartered within 50 miles from the focal institution. Because passing (relative to failing) a governance proposal makes a focal institution more likely to oppose management at its portfolio firms, the neighbors of the focal institution should become more likely to oppose management at their portfolio firms when the firms are heavily held by the focal institution. Continuing with the illustrative example, because institution A becomes more active in opposing management at its portfolio companies relative to institution B after the close-vote event, positive peer effects predict that, relative to institution B's neighbors, institution A's neighbors should become more active in opposing management at their portfolio firms when the firms are heavily held by institution A. In contrast, there should be little change in the voting behavior of institution A's neighbors at firms that are not held by institution A because institution A does not vote on proposals at such firms and hence, would have little influence on its neighbors' voting behavior at such firms.

I use a triple-difference specification to test this prediction. The first difference is between neighbors of focal institutions that pass a close vote and those of focal institutions that fail a close vote (*Pass*). The second

difference is from before to after the close-vote event (*Post*). The third difference is between stocks with a high exposure to the influence of the focal institution and those with a low exposure (*Exposure*). I use two measures to capture the exposure to the influence of the focal institution. The first is the fractional ownership of the focal institution in the stock, and the second is an indicator for whether the focal institution is a top 10 institutional shareholder of the stock. The triple-difference term ($Pass \times Post \times Exposure$) captures how the close-vote outcome (pass or fail) at a focal institution affects its neighbors' votes in stocks with differential exposure to the focal institution after the close-vote event than before. If positive peer effects exist in institutions' proxy voting behavior, the triple-difference term should be positive and significant.

I find evidence consistent with positive peer effects. Specifically, relative to the neighbors of a focal institution that rejects a close vote, the neighbors of a focal institution that passes a close vote become more likely to vote against the management of a firm after the close-vote event than before the event, when the firm is held more heavily by the focal institution. The economic magnitude of this effect is also significant. For instance, the coefficient on the triple-difference term is 0.8–1.1 percentage points when exposure is measured using the indicator for whether the focal institution is a top 10 institutional shareholder. This result is obtained after controlling for a host of fixed effects, including close-vote event fixed effects, proposal fixed effects, institution $\times$ year fixed effects, institution $\times$ proposal type $\times$ year fixed effects, institution $\times$ industry $\times$ year fixed effects, institution $\times$ headquarters state $\times$ year fixed effects, and institution $\times$ firm fixed effects.

The research design allows me to control for contextual and correlated effects to a large extent and eliminate many potential sources of omitted variable bias that can confound inferences. For example, the institution $\times$ year fixed effects control for time-varying voting preferences of each neighboring institution. These fixed effects also rule out the possibility that institutions in a given locality respond to common shocks by changing their voting behavior. The institution $\times$ proposal type $\times$ year fixed effects control for time-varying heterogeneity in an institution's tendency to oppose management across proposal types. The institution $\times$ firm fixed effects control for time-invariant tendency of a given institution to vote against management at a given firm.

For an alternative mechanism to explain the results, the mechanism has to satisfy three conditions. First, it must coincide systematically with the close-vote events and influence the voting behavior of focal and neighboring institutions in the same direction as the close-vote events. Second, it should differentially impact the voting behavior of a given neighboring institution at a given point in time across stocks (i.e., the impact should

increase with the focal institution's ownership in the stock at which the vote is being held). Third, the mechanism should not be absorbed by the various sets of fixed effects. These requirements significantly limit the scope for alternative interpretations of the results.

An institution's incentive to influence its peers should be particularly strong in contentious proposals when their votes are more pivotal in influencing the vote outcome. Consistent with this, I find that the peer effect is significant for votes on contentious proposals and becomes insignificant and much smaller in magnitude for votes on noncontentious proposals. These results provide suggestive evidence that peer influences in proxy voting are particularly pronounced when the focal institution and its neighbors are more likely to sway the outcome by voting together.

The key identifying assumption in my empirical strategy is that, absent the close-vote events at focal institutions, trends in voting behavior are the same between focal institutions that narrowly pass a governance proposal and those that narrowly reject one and between the neighbors of these two groups of focal institutions. I verify the parallel-trends assumption by conducting falsification tests that falsely assume close-vote events occur 12 months prior to the actual dates. The results show that focal institutions and their neighbors do not exhibit significant changes in voting behavior around these pseudoevent dates preceding the actual events, suggesting that the parallel trends assumption is likely to hold in my setting.

The peer effects in voting behavior at the institution level may have aggregate effects on actual vote outcomes. That is, the passage of a governance proposal at a focal institution can make its neighboring institutions more active in voting against management at firms that are heavily held by the focal institution, thereby increasing the likelihood that the management of such firms loses votes on governance proposals. To examine the effect of close-vote events on vote outcomes, I perform a triple-difference test on the sample of proposals. If close-vote events at a focal institution have aggregate effects on the vote outcome, the triple-difference term ($Pass \times Post \times Exposure$) should be positive and significant. I find evidence consistent with this prediction. The economic magnitude is large. For example, relative to stocks that do not have a focal institution as a top 10 institutional shareholder, those that do experience an increase of around 1.8–2.0 percentage points in the likelihood that management loses a vote when the focal institution passes a close vote, relative to when it fails one, after the close-vote event than before. This result suggests that peer influence can create social multiplier effects and have large impacts on vote outcomes.

The findings on peer influences in proxy voting have important policy implications. Mutual funds have a fiduciary duty to cast proxy votes in the best interests of

their shareholders. As Securities and Exchange Commission (SEC) Acting Chair Allison H. Lee (2021) put it, “the act of voting is itself a critical part of funds’ and advisers’ fiduciary obligations.” However, the amount of effort institutional shareholders put into voting decisions depends on the payoff to the decisions. If institutions vote independently without taking into consideration the voting decisions of other shareholders, they may have little incentive to engage in active voting because the probability that an institution casts a decisive vote is generally negligible. Therefore, regulations that aim to facilitate peer influence (e.g., by increasing the ease of coordination among shareholders) are likely to enhance institutional shareholders’ incentive to play an active role in corporate voting and exert influence on corporate governance. Moreover, the findings that close-vote events at focal institutions lead to changes in the voting behavior of their neighbors and affect actual vote outcomes suggest a social multiplier effect in proxy voting. Shocks to the voting behavior of one shareholder can transmit to nearby shareholders, leading to large effects on vote outcomes.

The rest of the paper is organized as follows. Section 2 reviews the related literature. Section 3 describes the data and summary statistics. Section 4 presents the empirical results, and Section 5 concludes.

2. Related Literature

This paper is related to three strands of the literature, with the first being the recent literature on the proxy voting behavior of institutional investors. Recent studies show that mutual funds’ proxy voting decisions are influenced by a host of factors, such as business and social connections to management (Davis and Kim 2007, Ashraf et al. 2012, Butler and Gurun 2012, Cvijanović et al. 2016, Calluzzo and Kedia 2019), the costs and benefits of active voting (Iliev and Lowry 2015), proxy advisers’ recommendations (Malenko and Shen 2016), capital gains lock-in (Dimmock et al. 2018b), and holdings in same-industry firms (He et al. 2019). Peer influence in proxy voting decisions is relatively underexplored. An important exception is the study by Matvos and Ostrovsky (2010), which instruments for peer funds’ votes in a firm’s director elections using the average vote of these funds in other firms’ director elections in the prior year and shows that a mutual fund’s tendency to oppose management increases with the likelihood that peer funds oppose management. My paper complements theirs by using plausibly exogenous shocks to the voting behavior of focal institutions to disentangle peer effects.³

Second, this paper is related to the literature on peer effects in economics and finance. Recent studies establish causal peer effects in various settings, including academic outcomes (Zimmerman 2003), criminal activities (Kling

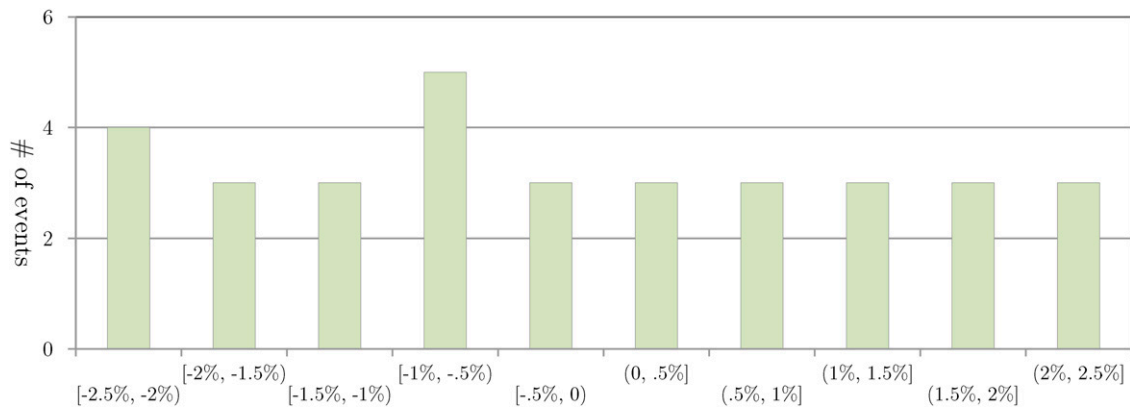
et al. 2005), corporate decision making (Shue 2013), the decision to become entrepreneurs (Lerner and Malmendier 2013), attitudes toward risk and trust (Ahern et al. 2014), financial misconduct (Dimmock et al. 2018a), and corporate social responsibility (Cao et al. 2019). My paper contributes to this literature in two important ways. First, it identifies peer effects in a setting characterized by strong strategic complementarities, namely proxy voting. Strategic complementarity generates an incentive to positively influence peers and can lead to large social multipliers (e.g., Glaeser et al. 2003). Second, compared with prior studies, the granular nature of the voting data enables me to include a rich set of fixed effects to disentangle peer effects from contextual and correlated effects.

Third, this paper also connects to the growing body of finance literature on geography. A number of studies suggest that, notwithstanding advances in communication technology, geographic distance still matters in financial decision making (e.g., monitoring by venture capitalists (Lerner 1995), stock investment decisions (Coval and Moskowitz 1999; 2001; Grinblatt and Keloharju 2001; Hong et al. 2005; Ivković and Weisbenner 2005; 2007; Brown et al. 2008), lending decisions (Petersen and Rajan 2002, Mian 2006, Sufi 2007), information production by financial analysts (Malloy 2005), payout policy (Becker et al. 2011), corporate investment decisions (Giroud 2013, Dougal et al. 2015), and financial misconduct (Parsons et al. 2018)). My paper adds to the literature by showing peer influences among geographically proximate institutions in proxy voting decisions. These findings suggest that geographic proximity might mitigate collective action problems inherent in the provision of monitoring effort by institutional investors.

3. Data and Summary Statistics

I identify publicly traded financial institutions by matching companies in the Institutional Shareholder Services (ISS) Voting Outcome data set with institutions in Thomson Reuters Institutional Holdings (13F) database during the period from 2003 through 2018. This matching procedure yields 66 institutions in the 13F database that are publicly traded or affiliated with a public company. To focus on votes at these publicly traded institutions for which the outcome is likely to be exogenous, I use a sample of close votes on shareholder-sponsored governance proposals (following Cuñat et al. 2012), defined as those that pass or fail within $\pm 2.5\%$ around the majority threshold.⁴ The outcome of nonclose votes, in contrast, could be driven by contextual or correlated effects. For example, a shareholder-sponsored governance proposal that passes or fails by a large margin at a publicly traded financial institution might be systematically related to common shocks that could affect all institutions in the locality (e.g., changes to state-level legislation concerning corporate governance). The final sample includes 33

Figure 1. (Color online) Distribution of Close-Vote Events at Publicly Traded Financial Institutions



Notes. This figure plots the number of shareholder-sponsored governance proposals at publicly traded financial institutions from 2003 through 2018 that pass or fail within $\pm 2.5\%$ around the majority threshold over 10 equally spaced bins, with a bin width of 0.5% .

close votes held at publicly traded financial institutions during the sample period, including 15 passing votes and 18 failing votes.⁵ Figure IA-1 in the online appendix plots the geographical distribution of the sample of close-vote events, with larger dots indicating a higher number of events. The events are generally scattered across the country.

Figure 1 shows the distribution of the sample of close-vote events over 10 equally spaced bins, with a bin width of 0.5% . The distribution of close votes at publicly traded financial institutions is smooth around the threshold. Consistent with Cuñat et al. (2012), a McCrary (2008) test fails to reject the null hypothesis of no manipulation. This result, however, should be viewed with caution for two reasons. First, the test may lack power given a small sample size. Second, as Bach and Metzger (2019) show, management can and often does manipulate close votes on shareholder proposals. Thus, an absence of manipulation may arise because managers of publicly traded financial institutions are not too opposed to these shareholder proposals or not sufficiently powerful to manipulate votes. Nevertheless, as long as the decision of whether to manipulate is not related to the expectation of the peer effects, which seems a reasonable assumption, the possibility of management manipulation of close votes and the information revealed by the outcome of these votes do not invalidate the identification of peer effects.

Table 1 shows the distribution of the close-vote events by year.⁶ There is relatively little clustering of close votes in particular years. Table 2 reports the distribution of the close votes by proposal type. The two most common proposal types are those that enhance shareholders' ability to call a special meeting and those advocating providing shareholders the right to act by written consent. Table 3 compares the pre-event characteristics of institutions that pass a close vote with those that fail a close vote. I regress various firm

characteristics, including the natural logarithm of the total assets (*Firm size*), the logarithm of the market-to-book ratio (*M/B*), Daniel, Grinblatt, Titman, and Wermers (DGTW 1997) characteristics-adjusted stock returns in the prior 12 months (*Abnormal stock return*), return on assets (*ROA*), book leverage (*Leverage*), and the fractional ownership by institutional investors in the firm (*Institutional ownership*), on an indicator for passing a close vote. The results show insignificant differences in these characteristics of institutions that fall on either side of the threshold, suggesting that the outcome of close votes is exogenous to the characteristics of the institutions and thus, can serve as plausibly exogenous shocks to the voting behavior of institutions.

I obtain data on mutual funds' votes on both management- and shareholder-sponsored governance proposals during the period from 2003 through 2018 from the ISS Voting Analytics database. The detailed voting information became available following the Securities and Exchange Commission's Rule 30b1-4 that requires all mutual funds registered in the United States to report their proxy votes in all shareholder meetings of their portfolio companies using Form N-PX starting from April 2003. For each proposed agenda item (i.e., proposal) voted by each mutual fund, the data report the firm that receives the proposal, the date of the shareholder meeting during which the proposal is considered, the issue being voted upon (e.g., board declassification, managerial compensation policies, or the elimination of poison pills), the sponsor of the proposal (i.e., management or shareholders), management's recommendation, the ISS recommendation, and the fund's vote (i.e., "for," "against," or "abstain").

I merge the voting data with the Thomson Reuters Institutional Holdings (13F) database. Because there is no common identifier for institutions across the two databases, I use a name-matching procedure to match the voting data with 13F. I identify 518 institutions in

Table 1. Close-Vote Events by Year

Year	No. of close votes	No. of passed votes	No. of failed votes	Average $ \text{margin} $, %
2003	3	1	2	1.416
2004	0	0	0	—
2005	3	0	3	0.56
2006	4	3	1	1.05
2007	3	3	0	1.05
2008	3	1	2	1.08
2009	5	1	4	1.10
2010	1	1	0	0.65
2011	4	2	2	1.72
2012	3	1	2	2.32
2013	0	0	0	—
2014	0	0	0	—
2015	1	0	1	1.07
2016	1	1	0	1.85
2017	0	0	0	—
2018	2	1	1	0.83
All years	33	15	18	1.33

Notes. This table reports the distribution of the close-vote events by year. Close-call votes are shareholder-sponsored governance proposals held at publicly traded financial institutions that pass or fail within $\pm 2.5\%$ around the majority threshold. *Average $|\text{margin}|$* is the average of the absolute value of the vote share margin.

the voting database that are 13F institutions. Mutual funds affiliated with these institutions cast about 96.8% of the votes covered by the voting database during the sample period.

The key outcome variable is institutions' tendency to vote against management recommendations. As a form of direct intervention ("voice") by shareholders, voting against management is widely used by institutional investors to engage with management of their portfolio firms (McCahery et al. 2016).⁷ Because mutual funds affiliated with the same fund company (i.e., institution) tend to vote in the same direction, I focus on the voting decision at the institution level. Following Cvijanović

et al. (2016) and He et al. (2019), I define the main dependent variable, $\text{VoteAgainstMgmt}_{i,p}$, as the number of funds affiliated with institution i that vote against management's recommendation on proposal p divided by the total number of funds affiliated with institution i voting on that proposal. Panel (A) of Table 4 shows that, in the full sample of votes by institutions on governance proposals, VoteAgainstMgmt has a mean of 0.125 and a standard deviation of 0.321.

I construct a sample of votes by the focal institutions on governance proposals during a 12-month period before and a 12-month period after the close-vote events. Panel (B) of Table 4 shows that VoteAgainstMgmt for

Table 2. Close-Vote Events by Proposal Type

Proposal type	No. of events
Amend articles/bylaws/charter—call special meetings	7
Provide right to act by written consent	5
Advisory vote to ratify named executive officers' compensation	4
Require a majority vote for the election of directors	4
Require independent board chairman	3
Declassify the board of directors	2
Restore or provide for cumulative voting	2
Eliminate or restrict severance agreements	1
Hire advisor/maximize shareholder value	1
Performance-based and/or time-based equity awards	1
Review executive compensation	1
Submit severance agreement (change in control) to shareholder vote	1
Submit shareholder rights plan (poison pill) to shareholder vote	1

Notes. This table reports the distribution of the close-vote events by proposal type. Close-call votes are shareholder-sponsored governance proposals held at publicly traded financial institutions that pass or fail within $\pm 2.5\%$ around the majority threshold.

Table 3. Pre-event Characteristics of Focal Institutions That Pass Vs. Fail a Close Vote

	Dependent					
	<i>Firm size</i> (1)	$\log(M/B)$ (2)	<i>Abn. stock return</i> (3)	<i>ROA</i> (4)	<i>Leverage</i> (5)	<i>Institutional ownership</i> (6)
<i>Pass</i>	−0.620 (0.73)	−0.063 (0.24)	0.002 (0.25)	−0.002 (0.17)	0.024 (0.40)	0.050 (1.00)
Constant	11.898 (16.42)***	0.503 (2.61)**	−0.005 (1.10)	0.023 (2.40)**	0.798 (18.76)***	0.678 (21.90)***
Observations	33	33	30	33	33	33
Adjusted R^2	0.02	0.00	0.00	0.00	0.01	0.03

Notes. This table compares the pre-event characteristics of focal institutions that pass a close vote with those of focal institutions that fail one. Close-call votes are shareholder-sponsored governance proposals held at publicly traded financial institutions that pass or fail within $\pm 2.5\%$ around the majority threshold. This table regresses various firm characteristics, including the natural logarithm of the total assets (*Firm size*), the logarithm of the market-to-book ratio (*M/B*), DGTW characteristics-adjusted stock returns in the prior 12 months (*Abnormal stock return*), return on assets (*ROA*), book leverage (*Leverage*), and the fractional ownership by institutional investors in the firm (*Institutional ownership*), on an indicator for passing a close vote. Numbers in parentheses are *t*-statistics based on robust standard errors.

Significance at the 5% level; *significance at the 1% level.

focal institutions has a mean of 0.130 and a standard deviation of 0.328.

I define neighboring institutions of a focal institution as those that are headquartered within 50 miles from the headquarters of the focal institution. To ensure that the results on peer effects are not contaminated by own effects, I exclude neighboring institutions that experience a close vote themselves during the 12 months before or 12 months after the close-vote event at the focal institution. I identify the location (zip code) of institutional investors by searching the filings by institutional investors on the SEC Edgar website.⁸ For a close-vote event at a focal institution, I retrieve the votes on governance proposals by the neighboring institutions of the focal institution during a 12-month window before and a 12-month window after the event. Panel (C) of Table 4 shows that *VoteAgainstMgmt* for the

neighboring institutions has a mean of 0.135 and a standard deviation of 0.328.

A stock's exposure to the influence of a focal institution varies with the focal institution's holdings in the stock because greater shareholding by the focal institution makes the institution's votes more pivotal. As such, the focal institution is more likely to influence its neighbors' voting behavior at a firm when the firm is heavily held by the focal institution than when it is not held. I use two measures to capture a stock's exposure to the influence of the focal institution. The first is the fractional ownership of the focal institution in the stock before the close-vote event (*FocalOwn*), and the second is an indicator for whether the focal institution is a top 10 institutional shareholder of the stock before the event (*FocalTop10*). Panel (C) of Table 4 shows that *FocalOwn* has a mean of 0.421% and a standard deviation of

Table 4. Summary Statistics of the Voting Sample

	Mean	Standard deviation	25th	50th	75th	No. of observations
Panel A: Full sample						
<i>VoteAgainstMgmt</i>	0.125	0.321	0.000	0.000	0.000	3,690,434
Panel B: Focal sample						
<i>VoteAgainstMgmt</i>	0.130	0.328	0.000	0.000	0.000	56,816
Panel C: Neighbor sample						
<i>VoteAgainstMgmt</i>	0.135	0.328	0.000	0.000	0.000	650,430
<i>FocalOwn</i> (%)	0.421	0.957	0.000	0.064	0.394	650,430
<i>FocalTop10</i>	0.067	0.250	0.000	0.000	0.000	650,430

Notes. This table presents the summary statistics for the sample of votes cast by institutions on governance proposals. The full sample consists of all votes on governance proposals by sample institutions. The focal sample consists of votes by focal institutions during a 12-month period before the close-vote event and a 12-month window after the event. The neighbor sample consists of votes by neighbors of focal institutions during a 12-month period before the close-vote event and a 12-month window after the event. Neighboring institutions are those that are headquartered within 50 miles from the headquarters of the focal institution and do not experience a close vote themselves during the 12 months before or 12 months after the close-vote event at the focal institution. *VoteAgainstMgmt* is the fraction of votes against management cast by funds affiliated with the institution. *FocalOwn* is the fraction of outstanding shares of the firm at which the proposal is being voted on that are held by the focal institution before the close-vote event. *FocalTop10* is an indicator that equals one if the focal institution is a top 10 institutional shareholder of the firm's stock before the close-vote event and zero otherwise. For each variable in each sample, I report the mean; standard deviation; and 25th, 50th, and 75th percentiles.

0.957%. The 25th percentile of *FocalOwn* is zero, suggesting that a substantial fraction of the observations has little exposure to treatment. *FocalTop10* has a mean of 0.067 and a standard deviation of 0.250.

4. Empirical Results

In this section, I first examine the effect of close-vote events on the voting behavior of focal institutions (the own effect) and on that of the neighbors of focal institutions (the peer effect). I then examine the effect of close-vote events on actual vote outcomes.

4.1. Own Effects

Although mutual funds have a fiduciary duty to cast proxy votes in the best interests of their shareholders, most fund managers do not have strong incentives to play an active role in voting because their primary job is to select and build portfolios and because they face collective action problems in their efforts to influence management. As a result, mutual funds may vote with management excessively. The passage of a governance proposal at a focal institution could make the institution more active in its voting decisions because of at least three reasons. First, the corporate governance team at an institution may infer the governance preferences of its shareholders from the outcome of their votes and align its voting decisions with the preferences of its shareholders.⁹ Thus, when a majority of an institution's shareholders vote for a governance proposal opposed by management, the institution may take a more proshareholder stance in its own voting decisions. Relatedly, the corporate governance team may be naturally more proactivist and feel emboldened when activist shareholders are more powerful in their own institution.

Second, the passage of a shareholder-sponsored governance proposal at an institution signals shareholder dissatisfaction with its management and corporate governance practices, which could prompt the institution to become more active on governance issues at its portfolio firms. According to former SEC commissioner Sommer (1992),¹⁰ "a substantial vote in favor of a shareholder proposal opposed by management might be seen as a vote of 'no confidence' in management and stir the juices of would-be 'raiders,' either proxy or takeover variety." Thus, losing a vote on governance proposals could induce management to take a more activist approach to corporate governance to appease shareholders.

Third, there could be a preach what you practice effect, in which institutions that adopt a governance practice cast votes in a manner consistent with that practice. Because the passage of a proposal on a governance policy at an institution increases the likelihood that the institution adopts the policy, the institution may become more likely to support proposals on similar governance policies at its portfolio companies. Commenting on mutual funds'

voting practices, Ralph Nader, a political activist, is quoted as saying that "[m]utual funds are a top-down autocracy. How can they be expected to go after the corporations they've invested in when there is such conformity of abuse? If they criticize executive compensation, they have to look at their own executive compensation" (*New York Times* 2016b).¹¹ Thus, the passage of a governance proposal at a focal institution may make the institution more active in promoting similar governance proposals at its portfolio firms.

The arguments suggest that the passage of a shareholder-sponsored governance proposal at a publicly traded financial institution may make the institution more active in voting against management at its portfolio firms. To examine the effects of close-vote events on focal institutions' voting behavior, I run the following difference-in-differences regression on the panel of focal institutions' votes on governance proposals at their portfolio companies around close-vote events,

$$\text{VoteAgainstMgmt}_{e,p} = \delta_e + \delta_p + \beta_1 \times \text{Pass}_e \times \text{Post}_{e,p} + \beta_2 \times \text{Post}_{e,p} + \varepsilon, \quad (1)$$

where $\text{VoteAgainstMgmt}_{e,p}$ is the fraction of votes against management cast by funds affiliated with the focal institution in event e on proposal p , δ_e is event fixed effects, δ_p is proposal fixed effects, Pass_e is an indicator that equals one if the focal institution in event e passes a close vote on a shareholder-sponsored governance proposal and zero if it rejects one, and $\text{Post}_{e,p}$ is an indicator that equals one if proposal p is in the 12-month period after event e and zero if it is in the 12-month period before the event. It is not necessary to include the Pass indicator in the regression because it is absorbed by the event fixed effects. I cluster standard errors at the close-vote event level to account for possible correlation in errors within events.¹² The coefficient on the interaction term (i.e., β_1) captures the treatment effect. If the passage (relative to rejection) of a shareholder-sponsored governance proposal at an institution makes the institution more active in monitoring its portfolio firms, the coefficient should be positive and significant.

I include close-vote event fixed effects and proposal fixed effects in all specifications. Close-vote event fixed effects control for any observed or unobserved event-specific heterogeneity. Proposal fixed effects control for heterogeneity across proposals (e.g., issues being voted on and recommendations by proxy advisors) as well as that across firms at which the proposals are being voted on (e.g., prior performance, ownership structure, information environment, and governance quality). For instance, some proposals may be associated with a high potential for agency conflicts and thus, require more intense scrutiny by institutions. A naïve comparison of institutions' votes across different proposals may capture differences in proposals rather than differences in voting behavior.

Table 5. The Effect of Close-Vote Events on the Voting Behavior of Focal Institutions

	Dependent variable = <i>VoteAgainstMgmt</i>		
	(1)	(2)	(3)
<i>Pass</i> × <i>Post</i>	0.024 (1.99)*	0.023 (2.52)**	0.027 (2.45)**
<i>Post</i>	−0.031 (1.46)	−0.033 (1.82)*	−0.029 (2.16)**
Event FEs	Yes	Yes	Yes
Proposal FEs	Yes	Yes	Yes
Institution × proposal type FEs	No	Yes	Yes
Institution × industry FEs	No	Yes	No
Institution × headquarters state FEs	No	Yes	No
Institution × firm FEs	No	No	Yes
Observations	34,747	34,725	33,698
Adjusted R^2	0.65	0.71	0.72

Notes. This table presents the results of the difference-in-differences tests of the impact of close-vote events on the voting behavior of focal institutions. The unit of observation is a focal institution's vote on a governance proposal. *VoteAgainstMgmt* is the fraction of votes against management cast by funds affiliated with the institution. *Pass* is an indicator that equals one if the focal institution passes a close vote and zero if it rejects one. *Post* is an indicator that equals one if the shareholder meeting at which the proposal is voted on is in the 12-month period after the close-vote event and zero if it is in the 12-month period before the event. Numbers in parentheses are *t*-statistics using standard errors clustered by close-vote event. FE, fixed effect.

*Significance at the 10% level; **significance at the 5% level.

The inclusion of proposal fixed effects enables me to rule out differences in proposals and time-varying firm attributes as potential explanations for the results. In some specifications, I further include institution × proposal type fixed effects, institution × industry fixed effects, institution × headquarters state fixed effects, and institution × firm fixed effects. These fixed effects rule out alternative interpretations based on time-invariant heterogeneity in an institution's tendency to oppose management across proposal types, industries, headquarters states, and firms.

Table 5 reports the results. The coefficient on the interaction term, β_1 , is positive and significant at conventional levels across all three specifications, indicating that focal institutions become more likely to oppose management at their portfolio companies after they pass a close vote on governance proposals. The economic magnitude of the own effect is nontrivial. Focal institutions that narrowly pass a shareholder-sponsored governance proposal become 2.3–2.7 percentage points more likely to oppose management after the close-vote event than before the event, relative to focal institutions that narrowly reject a shareholder-sponsored governance proposal.¹³ These results suggest that the passage of a shareholder-sponsored governance proposal at a focal institution makes the institution more active in voting against management at its portfolio companies.

4.2. Peer Effects

Because proxy voting exhibits strong strategic complementarities, institutions can attain a higher utility by conditioning their voting decisions on the voting behavior of their peers than by casting independent votes. Also, institutions may want to condition their votes on

those of their peers because of conformity concerns (e.g., Bernheim 1994) and/or because they expect reciprocal support from their peers on issues that are important to them. Institutions may coordinate their votes explicitly (e.g., by rallying support for their initiatives from their peers) or implicitly (e.g., by conditioning their voting decisions on their peers' votes).¹⁴ If institutions indeed coordinate their votes with their neighbors, the change in the voting behavior of focal institutions induced by close-vote events should lead to changes in the voting behavior of the neighboring institutions. Importantly, there should be variation in the impact of close-vote events across stocks within a given neighboring institution's portfolio at a given point in time. That is, a close-vote event at a focal institution should induce its neighbors to change their voting behavior at annual meetings of firms in which the focal institution holds relatively large equity stakes. At the other extreme, a close-vote event should have little impact on the neighbors' voting behavior at firms that are not held by the focal institution.

I use the panel of neighboring institutions' votes on governance proposals at their portfolio companies around close-vote events to examine the peer effects. The unit of observation is a neighboring institution's vote on a governance proposal. I run the following reduced-form triple-difference regression,¹⁵

$$\begin{aligned}
 \text{VoteAgainstMgmt}_{e,n,p} = & \delta_e + \delta_p + \delta_{n,t} + \gamma_1 \times \text{Pass}_e \\
 & \times \text{Post}_{e,p} \times \text{Exposure}_{ef} + \gamma_2 \\
 & \times \text{Pass}_e \times \text{Exposure}_{ef} + \gamma_3 \\
 & \times \text{Post}_{e,p} \times \text{Exposure}_{ef} + \gamma_4 \\
 & \times \text{Pass}_e \times \text{Post}_{e,p} + \gamma_5 \times \text{Post}_{e,p} \\
 & + \gamma_6 \times \text{Exposure}_{ef} + \varepsilon, \quad (2)
 \end{aligned}$$

where $VoteAgainstMgmt_{e,n,p}$ is the fraction of votes against management on proposal p cast by funds affiliated with institution n , which is a neighbor of the focal institution in close-vote event e ; δ_e is event fixed effects; δ_p is proposal fixed effects; $\delta_{n,t}$ is institution $\times$ year fixed effects; $Pass_e$ is an indicator that equals one if the focal institution in event e passes a close vote and zero if it rejects one; $Post_{e,p}$ is an indicator that equals one if proposal p is in the 12-month period after event e and zero if it is in the 12-month period before the event; and $Exposure_{ef}$ is either the fractional ownership of the focal institution in event e in firm f (i.e., the firm that receives proposal p) at the quarter-end before the close-vote event ($FocalOwn$) or an indicator for whether the focal institution in event e is a top 10 institutional shareholder of firm f at the quarter-end before the event ($FocalTop10$). I use the holdings of the focal institution in the

quarter immediately before the close-vote event to capture the exposure to treatment as this measure is plausibly exogenous to changes in neighboring institutions' voting behavior. I again cluster standard errors at the close-vote event level. The triple-difference term captures how close-vote outcomes (pass or fail) at a focal institution affect its neighbors' votes across stocks with differential exposure to the influence of the focal institution after the close-vote event than before. If positive peer effects exist in institutions' proxy voting behavior, the coefficient on the triple-difference term (i.e., γ_1) should be positive and significant.

Table 6 presents the regression results. I include close-vote event fixed effects and proposal fixed effects in all specifications. Columns (1) and (4) further include institution $\times$ year fixed effects to control for time-varying heterogeneity across neighboring institutions.

Table 6. The Effect of Close-Vote Events at a Focal Institution on the Voting Behavior of Its Neighbors

	Dependent variable = $VoteAgainstMgmt$					
	(1)	(2)	(3)	(4)	(5)	(6)
$Pass \times Post \times FocalOwn$	0.169 (2.40)**	0.261 (3.45)***	0.230 (2.57)**			
$Pass \times FocalOwn$	-0.041 (0.73)	-0.145 (4.10)***	-0.098 (1.59)			
$Post \times FocalOwn$	-0.130 (2.42)**	-0.099 (2.04)*	-0.054 (0.98)			
$FocalOwn$	0.056 (1.48)	0.056 (2.74)**	0.016 (0.41)			
$Pass \times Post \times FocalTop10$				0.008 (2.89)***	0.011 (3.42)***	0.008 (2.33)**
$Pass \times FocalTop10$				-0.000 (0.20)	-0.004 (2.81)***	-0.002 (1.05)
$Post \times FocalTop10$				-0.005 (2.21)**	-0.004 (1.54)	-0.003 (1.13)
$FocalTop10$				0.003 (2.10)**	0.001 (0.92)	0.000 (0.29)
$Pass \times Post$	-0.001 (0.43)	-0.002 (0.76)	-0.001 (1.10)	-0.001 (0.33)	-0.001 (0.65)	-0.001 (0.79)
$Post$	-0.004 (2.06)*	-0.001 (0.41)	-0.002 (0.87)	-0.004 (2.14)**	-0.001 (0.59)	-0.002 (0.93)
Proposal FEs	Yes	Yes	Yes	Yes	Yes	Yes
Event FEs	Yes	Yes	Yes	Yes	Yes	Yes
Institution $\times$ year FEs	Yes	No	Yes	Yes	No	Yes
Institution $\times$ proposal type $\times$ year FEs	No	Yes	No	No	Yes	No
Institution $\times$ industry $\times$ year FEs	No	Yes	No	No	Yes	No
Institution $\times$ headquarters state $\times$ year FEs	No	Yes	No	No	Yes	No
Institution $\times$ firm FEs	No	No	Yes	No	No	Yes
Observations	620,202	616,788	614,166	620,202	616,788	614,166
Adjusted R^2	0.58	0.74	0.64	0.58	0.74	0.64

Notes. This table presents the results of the triple-difference tests of the impact of close-vote events at a focal institution on the voting behavior of the institution's neighbors. The unit of observation is a neighboring institution's vote on a governance proposal. Neighboring institutions are those that are headquartered within 50 miles from the headquarters of the focal institution and do not experience a close vote themselves during the 12 months before or 12 months after the close-vote event at the focal institution. $Pass$ is an indicator that equals one if the focal institution passes a close vote and zero if it rejects one. $Post$ is an indicator that equals one if the shareholder meeting at which the proposal is voted on is in the 12-month period after the close-vote event and zero if it is in the 12-month period before the event. $FocalOwn$ is the fraction of outstanding shares of the firm at which the proposal is being voted on that are held by the focal institution before the close-vote event. $FocalTop10$ is an indicator that equals one if the focal institution is a top 10 institutional shareholder of the firm before the close-vote event and zero otherwise. Numbers in parentheses are t -statistics using standard errors clustered by close-vote event. FE, fixed effect.

*Significance at the 10% level; **significance at the 5% level; ***significance at the 1% level.

Columns (2) and (5) replace the institution $\times$ year fixed effects with institution $\times$ proposal type $\times$ year fixed effects, institution $\times$ industry $\times$ year fixed effects, and institution $\times$ headquarters state $\times$ year fixed effects. Columns (3) and (6) include institution $\times$ year fixed effects and institution $\times$ firm fixed effects. Consistently across all specifications, the coefficient on the triple-interaction term, γ_1 , is positive and significant at conventional levels, indicating that the passage of a close vote at a focal institution makes its neighbors more likely to oppose management of firms that are more heavily held by the focal institution.

In terms of the economic significance of the peer effect, columns (1)–(3) show that a one-standard deviation increase in the ownership by a focal institution (about 0.00957) increases the neighboring institutions' likelihood of opposing management by 0.162–0.250 percentage points when the focal institution passes a close vote, relative to when it rejects one, after the close-vote event than before. Similarly, columns (4)–(6) show that for stocks with a focal institution as a top 10 institutional shareholder, relative to those without, the neighboring institutions' likelihood of opposing management increases by 0.8–1.1 percentage points when the focal institution passes a close vote, relative to when it rejects one, after the close-vote event than before.

The extensive sets of fixed effects control for contextual and correlated effects to a large extent. The proposal fixed effects absorb any observed and unobserved heterogeneity across proposals as well as that across firms at which the proposals are being voted on. These fixed effects enable me to rule out differences in proposals and time-varying firm attributes as potential explanations for the results. For example, proxy advisors' recommendation on a proposal may lead to correlated votes by institutions that follow the recommendation. Such heterogeneity across proposals is soaked up by the proposal fixed effects.

With the institution $\times$ year fixed effects, the identification comes from variation across stocks with differential exposure to the focal institution within a given neighboring institution's portfolio in a given year. These fixed effects enable me to rule out alternative interpretations based on observed and/or unobserved time-varying heterogeneity across institutions. For example, changes in the general voting preferences of neighboring institutions because of factors such as changes in the amount of resources allocated to proxy research and shifts in investment horizons or objectives cannot explain the results. The institution $\times$ year fixed effects also rule out the possibility that the triple-difference results are driven by common shocks that simultaneously affect the voting behavior of focal institutions and that of their neighboring institutions. For example, the presence of a local governance advocacy group may happen to be correlated with the passage of

close votes at focal institutions and make both focal and neighboring institutions more likely to oppose management. Also, the passage of a close-vote event at a focal institution may increase the likelihood that neighboring institutions will pass similar proposals. Such common shocks will be subsumed by the institution $\times$ year fixed effects.

The specifications with the institution $\times$ proposal type $\times$ year fixed effects, institution $\times$ industry $\times$ year fixed effects, and institution $\times$ headquarters state $\times$ year fixed effects further rule out explanations based on time-varying heterogeneity in a neighboring institution's tendency to oppose management across proposal types, industries, and states where firms are headquartered, respectively. For instance, institutions may react differently to changes in the economic conditions of an industry (e.g., some institutions may increase engagement with firms in an industry when the industry performs poorly). Such time-varying heterogeneity is absorbed by the institution $\times$ industry $\times$ year fixed effects. Also, the institution $\times$ headquarters state $\times$ year fixed effects fully absorb differential responses of institutions in their voting behavior to changes at the state level that could affect firms headquartered in respective states (e.g., changes in state-level economic conditions and state law changes affecting corporate governance).

Furthermore, the institution $\times$ firm fixed effects control for time-invariant heterogeneity across neighboring institution-firm pairs in the tendency to vote against management. This forces identification to come from variation in exposure to focal institutions over time within a neighboring institution-firm pair. These fixed effects rule out alternative interpretations based on neighboring institutions' time-invariant voting behavior at given firms. For example, institutions may vote excessively with management at firms with which they have business or social ties (e.g., Butler and Gurun 2012, Cvijanović et al. 2016).

For an alternative mechanism to explain the results, the mechanism has to satisfy three conditions. First, it must coincide systematically with the close-vote events and influence the voting behavior of institutions in the same direction as the close-vote events. Second, for a given neighboring institution at a given point in time, the mechanism should have differential impacts on the institution's voting behavior across stocks in its portfolio. That is, the impacts should increase with the focal institution's ownership in the stock at which the vote is being held. Third, the mechanism should not be absorbed by the various sets of fixed effects. These requirements significantly limit the scope for alternative interpretations of the results.

I conduct a back-of-the-envelope calculation to estimate the size of a social multiplier. Because the average ownership of a focal institution in a stock with the focal institution as a top 10 institutional shareholder is 3.28%

and the average combined ownership of the sample neighboring institutions in such a stock is 4.41%, the last three columns of Table 6 indicate that close-vote events lead to an increase of 1.1–1.5 percentage points in the neighboring institutions' aggregate votes against management in stocks with the focal institution as a top 10 institutional shareholder (e.g., $0.011 \times \frac{4.41\%}{3.28\%} = 0.015$). Given that close-vote events increase the focal institution's likelihood of opposing management by 2.3–2.7 percentage points, these estimates imply a social multiplier ranging from 0.41 to 0.65. In other words, all else equal, for proposals at firms that have a focal institution as a top 10 institutional shareholder, a one-percentage point increase in the vote share against management because of against votes cast by the focal institution leads

to an increase of 0.41–0.65 percentage points in the vote share against management because of against votes by its neighboring mutual fund institutions. Because close-vote events at a focal institution can similarly influence the voting behavior of neighboring nonmutual fund institutions, which are not covered by the voting data set, these numbers likely represent a lower bound on the magnitude of the overall social multiplier.¹⁶

Peer influences in voting are likely to be particularly pronounced in contentious proposals on which individual shareholders' votes are more likely to matter. Put differently, a shareholder's incentive to influence its neighbors' voting decision should increase with the likelihood that the shareholders jointly could sway the outcome of the vote. To test this, I partition the sample into

Table 7. The Effect of Close-Vote Events at a Focal Institution on the Voting Behavior of Its Neighbors: Contentious Vs. Noncontentious Proposals

	Dependent variable = <i>VoteAgainstMgmt</i>							
	Contentious proposals				Noncontentious proposals			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
<i>Pass</i> × <i>Post</i> × <i>FocalOwn</i>	1.036 (2.54)**	1.468 (2.99)***			0.101 (1.38)	0.153 (1.87)*		
<i>Pass</i> × <i>FocalOwn</i>	−0.391 (1.40)	−0.488 (1.48)			−0.080 (1.61)	−0.112 (1.82)*		
<i>Post</i> × <i>FocalOwn</i>	−0.478 (1.69)	−0.589 (1.67)			−0.006 (0.17)	−0.020 (0.68)		
<i>FocalOwn</i>	0.252 (1.39)	0.295 (1.45)			0.001 (0.03)	0.004 (0.17)		
<i>Pass</i> × <i>Post</i> × <i>FocalTop10</i>			0.029 (2.47)**	0.045 (3.90)***			0.004 (1.41)	0.005 (1.53)
<i>Pass</i> × <i>FocalTop10</i>			−0.006 (0.65)	−0.004 (0.61)			−0.003 (1.41)	−0.004 (1.61)
<i>Post</i> × <i>FocalTop10</i>			−0.017 (1.82)*	−0.016 (1.80)*			−0.000 (0.29)	−0.001 (0.77)
<i>FocalTop10</i>			0.008 (1.30)	0.006 (1.19)			0.000 (0.18)	0.000 (0.25)
<i>Pass</i> × <i>Post</i>	−0.002 (0.87)	−0.005 (1.74)*	0.000 (0.15)	−0.003 (1.36)	−0.002 (1.05)	−0.001 (0.89)	−0.002 (1.00)	−0.001 (0.70)
<i>Post</i>	0.004 (1.23)	0.007 (1.93)*	0.003 (1.03)	0.005 (1.82)*	−0.000 (0.36)	−0.001 (1.03)	−0.000 (0.38)	−0.002 (1.05)
Proposal FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Event FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Institution × year FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Institution × firm FEs	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Observations	79,517	68,321	79,517	68,321	520,663	538,575	520,663	538,575
Adjusted R ²	0.70	0.74	0.70	0.74	0.41	0.61	0.41	0.61

Notes. This table presents the results of the triple-difference tests of the impact of close-vote events at a focal institution on neighboring institutions' voting behavior on contentious and noncontentious issues separately. The unit of observation is a neighboring institution's vote on a governance proposal. *Pass* is an indicator that equals one if the focal institution passes a close vote and zero if it rejects one. *Post* is an indicator that equals one if the shareholder meeting at which the proposal is voted on is in the 12-month period after the close-vote event and zero if it is in the 12-month period before the event. *FocalOwn* is the fraction of outstanding shares of the firm at which the proposal is being voted on that are held by the focal institution before the close-vote event. *FocalTop10* is an indicator that equals one if the focal institution is a top 10 institutional shareholder of the firm before the close-vote event and zero otherwise. The first (last) four columns use the subsample of contentious (noncontentious) proposals. Contentious proposals are defined as either those for which ISS recommends against management (the odd-numbered columns) or those that belong to proposal types for which the absolute value of the average margin of victory in the past three years is in the bottom tercile (the even-numbered columns), and noncontentious proposals are otherwise. Numbers in parentheses are *t*-statistics using standard errors clustered by close-vote event. FE, fixed effect.

*Significance at the 10% level; **significance at the 5% level; ***significance at the 1% level.

contentious proposals and noncontentious proposals. Contentious proposals are defined as either those for which ISS recommends against management (following Dimmock et al. 2018b) or those that belong to proposal types for which the absolute value of the average margin of victory in the past three years is in the bottom tercile, and noncontentious proposals are otherwise. About 10% of the sample is classified as contentious votes under each definition.

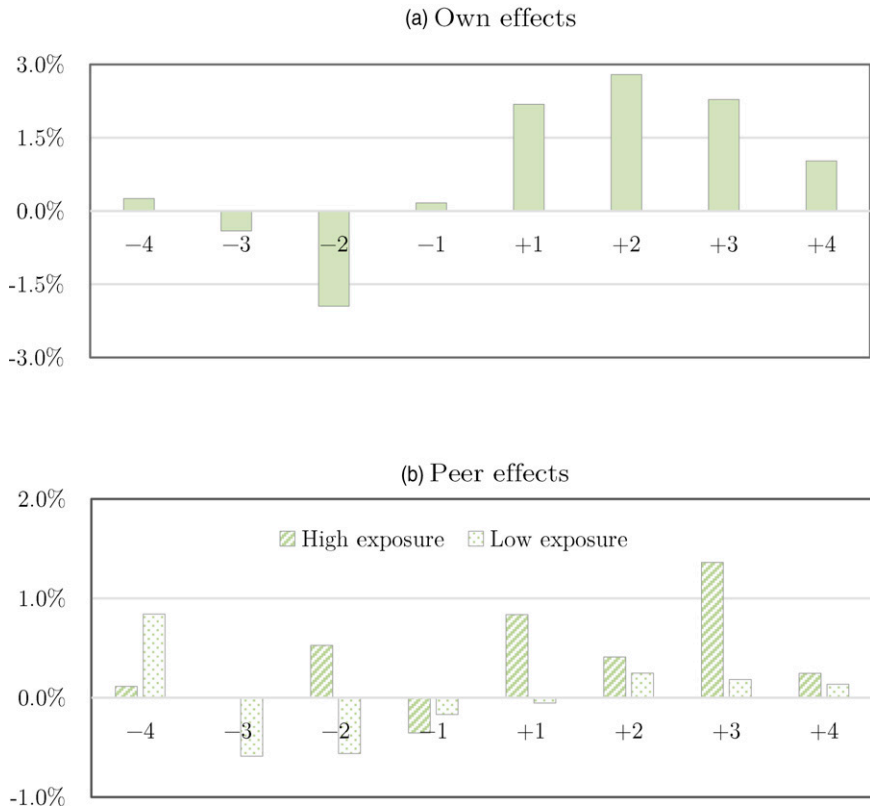
I run separate regressions on the two subsamples to shed light on the cross-sectional differences in peer effects. Table 7 reports the results using the most stringent specification with close-vote event fixed effects, proposal fixed effects, institution $\times$ year fixed effects, and institution $\times$ firm fixed effects. The coefficient estimate on the triple-interaction term for the subsample of votes on contentious proposals appears much larger in magnitude than that for the subsample of votes on noncontentious proposals. For example, columns (2) and (4) show that the coefficients on the triple-interaction term for

contentious proposals are 1.468 and 0.045, respectively, as compared with 0.153 and 0.005 for noncontentious proposals reported in columns (6) and (8), respectively. These results provide suggestive evidence that peer influences in monitoring are particularly pronounced when the focal institution and its neighbors are more likely to tip the balance in favor of their preferred outcomes.

4.3. Additional Tests

4.3.1. Graphical Evidence. Figure 2(a) plots the difference in the tendency to vote against management between focal institutions that narrowly pass a governance proposal and those that narrowly reject one during a 12-month period before and a 12-month period after the close-vote events. Each 12-month period is divided into four 3-month periods. Consistent with the results in Table 5, focal institutions that narrowly pass a shareholder-sponsored governance proposal, compared with those that narrowly fail one, become more likely to oppose management in voting after the close-vote event. For example,

Figure 2. (Color online) Graphical Representation of the Own and Peer Effects



Notes. This figure shows the own and peer effects induced by close-vote events. Panel (a) plots the difference in the tendency to vote against management between focal institutions that narrowly pass a governance proposal and those that narrowly reject one over a 12-month period before and a 12-month period after the close-vote events. Each 12-month period is divided into four 3-month periods. For example, the three-month period immediately following (before) the close-vote event is indicated by +1 (−1) on the horizontal line. Panel (b) plots the difference in the tendency to vote against management between neighbors of focal institutions that narrowly pass a governance proposal and neighbors of focal institutions that narrowly reject one during a 12-month period before and a 12-month period after the close-vote events separately for high-exposure (striped bars) and low-exposure (dotted bars) groups. The high-exposure group consists of votes cast by neighboring institutions at firms that have above-the-median fractional holdings by the focal institution, and the low-exposure group consists of the rest of the votes. I adjust an institution's tendency to vote against management on a proposal by subtracting the average tendency to vote against management by all institutions on that proposal and winsorize the variable at the top and bottom 5% to minimize the influence of outliers.

the difference in the tendency to vote against management is close to zero in the three months immediately before the event and increases to about 2.2 percentage points in the three months immediately after the event.

Figure 2(b) plots the difference in the tendency to vote against management between neighbors of focal institutions that narrowly pass a governance proposal and neighbors of focal institutions that narrowly reject one during a 12-month period before and a 12-month period after the close-vote events separately for high- and low-exposure groups. The high-exposure group consists of votes cast by neighboring institutions at firms that have above-the-median fractional holdings by the focal institution, and the low-exposure group consists of the rest of the votes. The graph shows that for the high-exposure group, neighbors of focal institutions that narrowly pass a governance proposal become more likely to oppose management in their votes than neighbors of focal institutions that narrowly reject a governance proposal, whereas for the low-exposure group, this change in the difference in voting behavior is much smaller. The graphical evidence is consistent with the results in Table 6, suggesting that the passage of a close vote at a focal institution makes its neighbors more likely to oppose management of firms that are more heavily held by the focal institution.

4.3.2. Pretrends. The identification strategy assumes that absent close-vote events at focal institutions, trends in voting behavior are the same between focal institutions that pass a close vote and those that reject one and between the neighbors of these two groups of focal institutions. To verify the parallel-trends assumption, I conduct falsification tests by assuming that the close-vote events occur 12 months prior to the actual dates. If the voting behavior of focal institutions exhibits parallel

trends, the change in the voting behavior of focal institutions that pass a close vote should not differ significantly from that of focal institutions that fail one around these “pseudoevents.” To test this, I repeat the regression in Equation (1) using the votes by focal institutions in the 12 months before the pseudoevents and the 12 months after. The results, reported in Table 8, show that the coefficient on the interaction term, $Pass \times Post$, is insignificant, suggesting that these two groups of focal institutions follow parallel trends in voting behavior in the two years prior to the actual events.

To test whether the neighbors of focal institutions behave similarly in the two years prior to the close-vote events, I repeat the regression in Equation (2) using the votes by neighboring institutions in the 12 months before the pseudoevents and the 12 months after. The results, reported in Table 9, show that the coefficient on the triple-difference term, $Pass \times Post \times Exposure$, is insignificant in all columns, suggesting that the neighbors of focal institutions that pass a close vote and those of focal institutions that reject one exhibit similar patterns in voting behavior in the two years prior to the actual events. These results suggest that the parallel trends assumption is likely to hold in my setting.

4.3.3. Nonneighboring Institutions. Nonneighbors of a focal institution, like neighbors, may benefit from conditioning their decisions to oppose management on those of the focal institution because doing so increases the likelihood that the outcome of the vote goes in their favor. However, because opposing management is privately costly, nonneighbors may deviate from a cooperative equilibrium and free ride on the monitoring effort of the focal institution when the costs of defection are low. Therefore, how a focal institution influences its distant

Table 8. Falsification Tests Using Pre-event Data: The Voting Behavior of Focal Institutions Around Pseudoevents

	Dependent variable = <i>VoteAgainstMgmt</i>		
	(1)	(2)	(3)
<i>Pass</i> × <i>Post</i>	0.008 (0.37)	0.021 (0.87)	0.005 (0.54)
<i>Post</i>	−0.002 (0.14)	0.004 (0.89)	−0.009 (1.15)
Event FEs	Yes	Yes	Yes
Proposal FEs	Yes	Yes	Yes
Institution × proposal type FEs	No	Yes	No
Institution × industry FEs	No	Yes	No
Institution × headquarters state FEs	No	Yes	No
Institution × firm FEs	No	No	Yes
Observations	35,502	35,388	34,285
Adjusted <i>R</i> ²	0.02	0.44	0.14

Notes. This table presents a falsification test assuming that the close-vote events are held 12 months before the actual date. The table tests the effect of the pseudoevents on the voting behavior of focal institutions. *Pass* is an indicator that equals one if the focal institution passes a close vote and zero if it rejects one. *Post* is an indicator that equals one if the shareholder meeting at which the proposal is voted on is in the 12-month period after the pseudoevent and zero if it is in the 12-month period before the event. Numbers in parentheses are *t*-statistics based on standard errors clustered at the close-vote event level. FE, fixed effect.

Table 9. Falsification Tests Using Pre-event Data: The Voting Behavior of the Neighbors of Focal Institutions Around Pseudoevents

	Dependent variable = <i>VoteAgainstMgmt</i>					
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Pass</i> × <i>Post</i> × <i>FocalOwn</i>	0.055 (0.23)	−0.010 (0.06)	−0.202 (1.37)			
<i>Pass</i> × <i>FocalOwn</i>	−0.030 (0.18)	−0.059 (0.46)	0.110 (1.15)			
<i>Post</i> × <i>FocalOwn</i>	−0.187 (2.13)**	−0.077 (1.45)	−0.065 (1.18)			
<i>FocalOwn</i>	0.115 (1.89)*	0.028 (0.83)	0.039 (1.28)			
<i>Pass</i> × <i>Post</i> × <i>FocalTop10</i>				0.003 (0.48)	−0.002 (0.59)	−0.003 (0.48)
<i>Pass</i> × <i>FocalTop10</i>				0.000 (0.08)	0.001 (0.32)	0.003 (1.04)
<i>Post</i> × <i>FocalTop10</i>				−0.005 (1.48)	−0.000 (0.11)	−0.002 (0.84)
<i>FocalTop10</i>				0.005 (2.10)**	−0.001 (0.58)	0.001 (0.88)
<i>Pass</i> × <i>Post</i>	0.005 (0.83)	0.004 (0.95)	0.003 (0.91)	0.005 (0.91)	0.004 (1.08)	0.003 (0.83)
<i>Post</i>	0.001 (0.19)	0.003 (0.63)	0.001 (0.25)	0.001 (0.09)	0.002 (0.53)	0.001 (0.20)
Proposal FEs	Yes	Yes	Yes	Yes	Yes	Yes
Event FEs	Yes	Yes	Yes	Yes	Yes	Yes
Institution × year FEs	Yes	No	Yes	Yes	No	Yes
Institution × proposal type × year FEs	No	Yes	No	No	Yes	No
Institution × industry × year FEs	No	Yes	No	No	Yes	No
Institution × HQ state × year FEs	No	Yes	No	No	Yes	No
Institution × firm FEs	No	No	Yes	No	No	Yes
Observations	524,690	520,884	518,325	524,690	520,884	518,325
Adjusted R ²	0.58	0.74	0.63	0.58	0.74	0.63

Notes. This table presents a falsification test assuming that the close-vote events are held 12 months before the actual date. The table tests the effect of the pseudoevents on the voting behavior of the neighbors of focal institutions. *Pass* is an indicator that equals one if the focal institution passes a close vote and zero if it rejects one. *Post* is an indicator that equals one if the shareholder meeting at which the proposal is voted on is in the 12-month period after the pseudoevent and zero if it is in the 12-month period before the event. *FocalOwn* is the fraction of outstanding shares of the firm at which the proposal is being voted on that are held by the focal institution before the pseudoevent. *FocalTop10* is an indicator that equals one if the focal institution is a top 10 institutional shareholder of the firm before the pseudoevent and zero otherwise. Numbers in parentheses are *t*-statistics based on standard errors clustered at the close-vote event level. FE, fixed effect; HQ, headquarters.

*Significance at the 10% level; **significance at the 5% level.

peers in voting depends on the relative strength of the coordination incentive and the free-riding incentive.

To examine the effect of close-vote events at a focal institution on the voting behavior of nonneighbors, I classify nonneighbors into three groups with increasing distance from the focal institution. The first group consists of institutions that are headquartered between 50 and 200 miles from the focal institution. The second group consists of institutions that are headquartered between 200 and 500 miles from the focal institution. The third group consists of institutions that are more than 500 miles away from the focal institution. For each group, I exclude peer institutions that experience a close vote themselves during the 12 months before or 12 months after the close-vote event at the focal institution.

Table 10 repeats the regression in Equation (2) for each group of nonneighbors separately. Panel (A) shows

that there is some evidence that close-vote events positively influence the voting behavior of the first group of nonneighbors (i.e., institutions that are headquartered between 50 and 200 miles from the focal institution). For example, the coefficient on the triple-interaction term is positive and significant at conventional levels in four of six specifications. In contrast, the results for the other two groups of nonneighbors, reported in panels (B) and (C), are generally insignificant. These results provide suggestive evidence that the coordination incentive and the free-riding incentive largely offset each other for nonneighboring institutions that are farther away. Because geographic distance hinders interactions among investors, the null results in panels (B) and (C) are also consistent with focal institutions having little influence on their geographically distant peers because of reduced opportunities to interact with each other.

Table 10. The Effect of Close-Vote Events at a Focal Institution on the Voting Behavior of Nonneighbors

	Dependent variable = <i>VoteAgainstMgmt</i>					
	(1)	(2)	(3)	(4)	(5)	(6)
Panel A: $50 < \text{distance} \leq 200$						
<i>Pass</i> × <i>Post</i> × <i>FocalOwn</i>	0.138 (1.93)*	0.046 (0.93)	0.006 (0.09)			
<i>Pass</i> × <i>Post</i> × <i>FocalTop10</i>				0.010 (4.14)***	0.004 (1.94)*	0.006 (2.05)*
Observations	920,410	916,753	910,222	920,410	916,753	910,222
Adjusted R^2	0.54	0.67	0.60	0.54	0.67	0.60
Panel B: $200 < \text{distance} \leq 500$						
<i>Pass</i> × <i>Post</i> × <i>FocalOwn</i>	−0.144 (1.61)	−0.022 (0.37)	0.002 (0.03)			
<i>Pass</i> × <i>Post</i> × <i>FocalTop10</i>				−0.005 (1.27)	−0.001 (0.46)	0.000 (0.01)
Observations	935,159	930,494	922,869	935,159	930,494	922,869
Adjusted R^2	0.59	0.74	0.66	0.59	0.74	0.66
Panel C: $\text{Distance} > 500$						
<i>Pass</i> × <i>Post</i> × <i>FocalOwn</i>	0.038 (1.41)	−0.018 (0.85)	0.016 (0.72)			
<i>Pass</i> × <i>Post</i> × <i>FocalTop10</i>				−0.000 (0.22)	−0.001 (1.09)	−0.000 (0.37)
Observations	4,068,421	4,064,458	4,062,361	4,068,421	4,064,458	4,062,361
Adjusted R^2	0.56	0.71	0.64	0.56	0.71	0.64
Proposal FEs	Yes	Yes	Yes	Yes	Yes	Yes
Event FEs	Yes	Yes	Yes	Yes	Yes	Yes
Institution × year FEs	Yes	No	Yes	Yes	No	Yes
Institution × proposal type × year FEs	No	Yes	No	No	Yes	No
Institution × industry × year FEs	No	Yes	No	No	Yes	No
Institution × HQ state × year FEs	No	Yes	No	No	Yes	No
Institution × firm FEs	No	No	Yes	No	No	Yes

Notes. This table presents the results of the triple-difference tests of the impact of close-vote events at a focal institution on the voting behavior of nonneighbors. The unit of observation is a nonneighbor's vote on a governance proposal. I classify nonneighbors into three groups: those that are headquartered between 50 and 200 miles from the focal institution (panel (A)), those between 200 and 500 miles (panel (B)), and those more than 500 miles away (panel (C)). *Pass* is an indicator that equals one if the focal institution passes a close vote and zero if it rejects one. *Post* is an indicator that equals one if the shareholder meeting at which the proposal is voted on is in the 12-month period after the close-vote event and zero if it is in the 12-month period before the event. *FocalOwn* is the fraction of outstanding shares of the firm at which the proposal is being voted on that are held by the focal institution before the close-vote event. *FocalTop10* is an indicator that equals one if the focal institution is a top 10 institutional shareholder of the firm before the close-vote event and zero otherwise. Numbers in parentheses are *t*-statistics using standard errors clustered by close-vote event. FE, fixed effect; HQ, headquarters.

*Significance at the 10% level; ***significance at the 1% level.

4.4. Vote Outcomes

Given the significant influence of close-vote events at a focal institution on the voting behavior of its neighbors, such events may have an impact on actual vote outcomes.

I use a triple-difference specification similar to Equation (2) to examine this question. The unit of observation in this analysis is a proposal. There are 148,344 proposals voted on by the neighboring institutions during the 12 months before and 12 months after the close-vote events. The dependent variable is an indicator that equals one if management loses the vote on a proposal and zero otherwise. Following Dimmock et al. (2018b), I control for various firm characteristics, including the natural logarithm of total assets (*Firm size*), market-to-book ratio (*M/B*), industry-adjusted stock returns in the prior 12 months (*Abnormal stock return*), book leverage

(*Leverage*), capital expenditures over assets (*CapEx*), whether the firm is included in the Standard and Poor's (S&P) 500 index (*SP500*), and the fractional ownership by institutional investors in the firm (*Institutional ownership*). I include a rich set of fixed effects to control for observed and unobserved heterogeneity, including close-vote event fixed effects, firm fixed effects, proposal type × year fixed effects, industry × year fixed effects, and headquarters state × year fixed effects. I again cluster standard errors at the close-vote event level to account for possible correlation in errors within events. The triple-interaction term (*Pass* × *Post* × *Exposure*) captures the effect of passing a close vote at a focal institution, relative to rejecting one, on the actual vote outcome of proposals across stocks with differential exposure to the focal institution after the close-vote event than before. If peer effects induced by close-vote

Table 11. The Effect of Close-Vote Events on Vote Outcomes

	Dependent variable = <i>Management loses</i>					
	(1)	(2)	(3)	(4)	(5)	(6)
<i>Pass</i> × <i>Post</i> × <i>FocalOwn</i>	0.528 (3.36)***	0.570 (3.92)***	0.449 (3.00)***			
<i>Pass</i> × <i>FocalOwn</i>	−0.327 (3.14)***	−0.276 (2.92)***	−0.202 (2.51)**			
<i>Post</i> × <i>FocalOwn</i>	−0.225 (1.88)*	−0.288 (2.43)**	−0.214 (1.88)*			
<i>FocalOwn</i>	0.125 (2.22)**	0.183 (2.62)**	0.078 (1.21)			
<i>Pass</i> × <i>Post</i> × <i>FocalTop10</i>				0.020 (3.47)***	0.020 (4.00)***	0.017 (3.01)***
<i>Pass</i> × <i>FocalTop10</i>				−0.011 (4.14)***	−0.011 (3.15)***	−0.009 (2.55)**
<i>Post</i> × <i>FocalTop10</i>				−0.008 (1.51)	−0.009 (2.01)*	−0.009 (1.73)*
<i>FocalTop10</i>				0.003 (1.32)	0.006 (1.91)*	0.004 (1.02)
<i>Pass</i> × <i>Post</i>	0.000 (0.14)	0.000 (0.00)	0.000 (0.02)	0.001 (0.41)	0.001 (0.37)	0.001 (0.26)
<i>Post</i>	0.002 (1.04)	0.001 (0.73)	−0.002 (0.67)	0.002 (0.86)	0.001 (0.41)	−0.002 (0.82)
<i>ISSWithMgmt</i>	−0.217 (17.01)***	−0.228 (17.96)***	−0.126 (22.08)***	−0.217 (17.02)***	−0.228 (17.96)***	−0.126 (22.08)***
<i>Firm size</i>	−0.002 (3.32)***	0.006 (4.02)***	−0.001 (0.57)	−0.002 (3.29)***	0.006 (4.04)***	−0.001 (0.55)
<i>M/B</i>	−0.000 (1.84)*	−0.000 (1.25)	−0.000 (1.05)	−0.000 (1.83)*	−0.000 (1.21)	−0.000 (1.03)
<i>Abnormal stock return</i>	−0.005 (3.70)***	−0.004 (5.27)***	−0.005 (5.40)***	−0.005 (3.71)***	−0.004 (5.26)***	−0.005 (5.38)***
<i>Leverage</i>	−0.003 (2.16)**	−0.001 (0.34)	−0.001 (0.41)	−0.003 (2.17)**	−0.001 (0.33)	−0.001 (0.42)
<i>CapEx</i>	0.014 (1.46)	−0.049 (2.55)**	−0.061 (3.83)***	0.014 (1.45)	−0.049 (2.54)**	−0.061 (3.86)***
<i>S&P 500</i>	0.026 (13.39)***	0.015 (3.60)***	0.001 (0.20)	0.026 (13.38)***	0.015 (3.60)***	0.001 (0.20)
<i>Institutional ownership</i>	0.031 (8.61)***	0.001 (0.27)	−0.000 (0.01)	0.031 (8.65)***	0.002 (0.33)	−0.000 (0.04)
Event FEs	Yes	Yes	Yes	Yes	Yes	Yes
Firm FEs	No	Yes	Yes	No	Yes	Yes
Proposal type × year FEs	No	No	Yes	No	No	Yes
Industry × year FEs	No	No	Yes	No	No	Yes
HQ state × year FEs	No	No	Yes	No	No	Yes
Observations	147,091	146,999	146,918	147,091	146,999	146,918
Adjusted R^2	0.18	0.25	0.49	0.18	0.25	0.49

Notes. This table presents the results of the triple-difference tests of the impact of close-vote events on vote outcomes. The unit of observation is a governance proposal. The dependent variable is an indicator that equals one if management loses the vote on a proposal and zero otherwise. *Pass* is an indicator that equals one if the focal institution passes a close vote and zero if it rejects one. *Post* is an indicator that equals one if the shareholder meeting at which the proposal is voted on is in the 12-month period after the close-vote event and zero if it is in the 12-month period before the event. *FocalOwn* is the fraction of outstanding shares of the firm at which the proposal is being voted on that are held by the focal institution before the close-vote event. *FocalTop10* is an indicator that equals one if the focal institution is a top 10 institutional shareholder of the firm before the close-vote event and zero otherwise. *ISSWithMgmt* is an indicator that equals one if ISS and management recommendations are the same and zero otherwise. I control for various firm characteristics, including the natural logarithm of the total assets (*Firm size*), market-to-book ratio (*M/B*), industry-adjusted stock returns in the prior 12 months (*Abnormal stock return*), book leverage (*Leverage*), capital expenditures over assets (*CapEx*), whether the firm is included in the S&P 500 index (*S&P 500*), and the fractional ownership by institutional investors in the firm (*Institutional ownership*). Numbers in parentheses are *t*-statistics using standard errors clustered by close-vote event. FE, fixed effect; HQ, headquarters.

*Significance at the 10% level; **significance at the 5% level; ***significance at the 1% level.

events have aggregate effects on the vote outcome, the coefficient on the triple-interaction term should be positive and significant.

The results, reported in Table 11, show that the coefficient on the triple-interaction term is positive and significant at the 1% level in all specifications. The

economic magnitude is large as well. The first three columns show that a one-standard deviation increase in the ownership by a focal institution (about 0.00957) increases the likelihood that management loses a vote by 0.430–0.545 percentage points when the focal institution passes a close vote, relative to when it rejects one, after the close-vote event than before. Similarly, the last three columns show that relative to stocks that do not have a focal institution as a top 10 institutional shareholder, those that do experience an increase of 1.8–2.0 percentage points in the likelihood that management loses a vote when the focal institution passes a close vote, relative to when it fails one, after the close-vote event than before. These magnitudes are sizable given that the standard deviation of the probability that management loses a proxy vote is 17.6 percentage points.

Table 11 also shows that consistent with prior studies (e.g., Malenko and Shen 2016), management is less likely to lose a vote when ISS recommends voting with management. The likelihood of management losing a vote is about 12.6–22.8 percentage points lower when ISS and management recommendations are the same than when they are different. Moreover, management is more likely to lose a vote when their firm's stock underperforms the industry, suggesting that poor performance leads to increased shareholder monitoring (e.g., Gillan and Starks 2000).

I use a similar triple-difference specification to test the impact of peer effects induced by close-vote events on abnormal stock returns around votes on governance proposals. The dependent variable is the abnormal stock return on the day of the vote on a governance proposal. Because a large majority of governance proposals pass or fail by a large margin, the stock return around the vote on such proposals provides little information about the valuation impact of the vote outcome. I thus restrict the sample for this analysis to governance proposals that pass or fail within $\pm 2.5\%$ around the majority threshold. Table IA-3 in the online appendix shows that the triple-interaction term is positive and significant at conventional levels in five of six specifications, suggesting that for close-call votes on governance proposals, an increase in voting against management induced by peer effects is associated with higher abnormal stock returns. To the extent that the stock return around close votes reflects the direct value of the proposals for shareholders, this finding is consistent with the view that opposing management on contentious governance issues is value enhancing (e.g., Alexander et al. 2010, Cuñat et al. 2012, Dimmock et al. 2018b). Thus, an increase in voting against management induced by peer effects is not only associated with a higher likelihood of management losing a vote but is also associated with higher abnormal stock returns when the vote is close.

5. Conclusion

This paper examines peer effects in proxy voting decisions, which are characterized by strong strategic complementarities. My identification strategy exploits plausibly exogenous variation in the voting behavior of publicly traded financial institutions induced by close-vote events at these institutions. I first show that passing a close vote on a shareholder-sponsored governance proposal, relative to rejecting one, makes a focal institution more likely to oppose management in its voting decisions.

Using a triple-difference approach, I find evidence consistent with positive peer effects. Specifically, relative to the neighbors of a focal institution that rejects a close vote, the neighbors of a focal institution that passes a close vote become more likely to vote against the management of a portfolio firm after the close-vote event than before when the firm is more heavily held by the focal institution. This effect is particularly strong when the proposal up for a vote is contentious. The granular voting data enable me to disentangle peer effects from contextual and correlated effects to a large extent by controlling for a rich set of fixed effects, including close-vote event fixed effects, proposal fixed effects, institution $\times$ proposal type $\times$ year fixed effects, institution $\times$ industry $\times$ year fixed effects, institution $\times$ headquarters state $\times$ year fixed effects, and institution $\times$ firm fixed effects. The triple-difference approach and the extensive sets of fixed effects substantially narrow the scope for alternative explanations of the results.

This paper contributes to our understanding of peer influence in the provision of monitoring effort by institutional investors. Because of strategic complementarities in proxy voting decisions, institutions have a strong incentive to influence their peers to make their aggregate votes more pivotal. Thus, regulatory policies that aim at facilitating peer influence may encourage more active participation of institutional investors in corporate voting. Moreover, the findings that the voting behavior of an institution positively influences that of its neighboring institutions suggest that there is a significant social multiplier effect. Shocks to the voting behavior of one shareholder can transmit to peer shareholders, leading to large aggregate effects on governance outcomes.

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and suggestions. The author retains responsibility for any remaining errors.

Endnotes

¹ For example, according to a recent survey of institutional investors conducted by Morrow Sodali (2018), a consulting firm specializing in shareholder services, 59% of the respondents, managing a combined \$13 trillion in assets, indicate that they would collaborate with other shareholders with respect to important issues at annual shareholder meetings. Also, according to Katherine Rabin (2013), the chief executive officer (CEO) of Glass Lewis (a major proxy advisor for institutional investors), many of its clients “across all investor types are now engaging with U.S. companies directly or in collaboration with other investors.”

² Asset managers in the same locality have increased opportunities for face-to-face interactions with one another (e.g., through “idea dinners” (New York Times 2005), industry conferences, and networking events hosted by local financial professional communities, such as Chartered Financial Analyst (CFA) chapters, belonging to the same private clubs, etc.). Consistent with the idea that geographic proximity facilitates sharing of value-relevant information among portfolio managers, Iliev and Lowry (2015) find that mutual funds located in cities with a high concentration of funds are more active in their voting decisions. Hong et al. (2005) and Pool et al. (2015) show that mutual fund managers located close by tend to place correlated trades and that such trades deliver abnormal returns.

³ More broadly, my paper contributes to the literature on the governance role of institutional investors by highlighting peer effects as an important channel through which institutional investors influence corporate governance. An incomplete list of empirical studies in this literature includes Brickley et al. (1988), Agrawal and Mandelker (1990), Gillan and Starks (2000), Huson et al. (2001), Parrino et al. (2003), Almazan et al. (2005), Gaspar et al. (2005), Chen et al. (2007), Matvos and Ostrovsky (2008), Aggarwal et al. (2011), Harford et al. (2011), Aghion et al. (2013), Appel et al. (2016; 2018), and Iliev et al. (2021).

⁴ The choice of a bandwidth of 2.5% around the majority threshold reflects a trade-off between selecting a bandwidth large enough to have a reasonable sample size to detect the effects yet small enough to avoid incorporating votes for which the outcome is likely to be endogenous to the characteristics of the institutions. Table IA-1 in the online appendix shows that the results are qualitatively similar when using a bandwidth of 2% or 3% around the majority threshold.

⁵ All of the shareholder-sponsored governance proposals involved in these events are opposed by management. Thus, the passage of a close-call shareholder-sponsored governance proposal at a focal institution means that a majority of shareholders vote against management’s recommendation on the proposal.

⁶ The drop in the number of close-vote events in more recent years is because of a decrease in the number of shareholder-sponsored governance proposals filed with publicly traded financial institutions rather than a decrease in the contentiousness of the proposals.

⁷ Voting against management recommendations does not necessarily imply that the votes are in the direction of increasing shareholder value because whether management’s recommendations increase or decrease shareholder value is unclear ex ante. Nevertheless, because opposing management is costly to institutions (e.g., Davis and Kim 2007, Cvijanović et al. 2016), doing so indicates that institutional investors take an active stance on voting.

⁸ Following previous studies (e.g., Baik et al. 2010), I use the headquarters location of institutional investors. Although some institutions have branches/offices in multiple locations, they typically have an in-house corporate governance team that advises on governance issues of portfolio companies.

⁹ Anecdotal evidence suggests that shareholders of asset management firms care about how their firms vote on corporate governance issues at portfolio firms. For example, a shareholder-sponsored proposal filed with BlackRock called for the asset management firm to evaluate “options for bringing its voting practices in line with its stated principle of linking executive compensation and performance.” The proposal criticizes BlackRock as “exercising less fiduciary responsibility—by a factor of ten—than the median investment manager” (see <https://www.sec.gov/Archives/edgar/data/1086462/000121465916010903/1415162px14a6g.htm>, accessed on April 20, 2020). Shareholders’ dissatisfaction with asset managers’ proxy voting records has attracted intense media attention (e.g., Bloomberg 2016, Fortune 2016, New York Times 2016a, The Nation 2016). Shareholders have also engaged asset management firms on their proxy voting policies on environmental, social, and governance (ESG) issues (e.g., Bloomberg 2019).

¹⁰ Del Guercio and Hawkins (1999) find that the passage of shareholder proposals leads to an increase in the frequency of proxy contest attempts.

¹¹ For example, a Financial Times (2016) article suggests that one reason for BlackRock’s soft stance on CEO pay at its portfolio companies is that BlackRock itself was “deficient in aligning pay with performance.” Similarly, former congressman Barney Frank expressed concerns that fund companies with highly paid executives “make poor overseers of CEO pay” (Reuters 2016).

¹² Table IA-2 in the online appendix shows that the results are robust to alternative clustering of standard errors, including two-way clustering by close-vote event and by institution-year, two-way clustering by close-vote event and by firm-shareholder meeting, and the wild bootstrap of Cameron et al. (2008) to correct for a small number of clusters.

¹³ The coefficient on *Pass × Post* appears larger in magnitude and more significant in column (3) with the institution × firm fixed effects. This could arise for two reasons. First, the inclusion of the institution × firm fixed effects forces identification to come from variation across institution-firm pairs in which the institution has a relatively long investment horizon in the stock, which could make the institution more active in voting (Iliev and Lowry 2015). Second, the institution × firm fixed effects can explain a significant portion of the variation in a focal institution’s tendency to vote against management across firms. Thus, their inclusion reduces residual variation and leads to more precise coefficient estimates on the own effect. Nevertheless, the proposals that provide identification in column (3) have statistically the same characteristics as the rest of the proposals, including the likelihood that ISS recommendations align with those of management and the likelihood of management losing a vote.

¹⁴ A series of regulatory reforms has significantly increased the ability and incentives of institutional investors to coordinate their votes at shareholder meetings. In particular, the 1992 proxy reform enables shareholders to communicate with one another as long as they do not directly seek the power to vote as proxy for other shareholders. Rule 30b1-4, adopted in 2003, mandates that registered mutual funds disclose “their actual voting records.” The disclosure makes defection from cooperative voting agreements readily detectable, thereby lowering enforcement costs. These reforms enhance the ability and willingness of institutional shareholders to act collectively to influence corporate management.

¹⁵ To test for peer effects, it is not sufficient to compare the change in the correlation of votes between a focal institution and its neighbors around close-vote events because an important source of identification comes from proposals that are voted on by the neighboring institutions but not by the focal institution.

¹⁶ In particular, the average combined fractional ownership by neighboring nonmutual fund institutions in stocks with the focal institution as a top 10 institutional shareholder is 5.14% as compared

with 4.46% by neighboring mutual fund institutions covered in the voting database. If the peer effect is similar across nonmutual fund and mutual fund institutions, the overall social multiplier would be much larger.

References

- Aggarwal R, Erel I, Ferreira M, Matos P (2011) Does governance travel around the world? Evidence from institutional investors. *J. Financial Econ.* 100(1):154–182.
- Aghion P, Van Reenen J, Zingales L (2013) Innovation and institutional ownership. *Amer. Econom. Rev.* 103(1):277–304.
- Agrawal A, Mandelker GN (1990) Large shareholders and the monitoring of managers: The case of antitakeover charter amendments. *J. Financial Quant. Anal.* 25(2):143–161.
- Ahern K, Ran D, Shumway T (2014) Peer effects in risk aversion and trust. *Rev. Financial Stud.* 27(11):3213–3240.
- Alexander C, Chen M, Seppi D, Spatt C (2010) Interim news and the role of proxy voting advice. *Rev. Financial Stud.* 23(12):4419–4454.
- Almazan A, Hartzell JC, Starks LT (2005) Active institutional shareholders and costs of monitoring: Evidence from executive compensation. *Financial Management* 34(4):5–34.
- Appel I, Gormley TA, Keim DB (2016) Passive investors, not passive owners. *J. Financial Econ.* 121(1):111–141.
- Appel I, Gormley TA, Keim DB (2018) Standing on the shoulders of giants: The effect of passive investors on activism. *Rev. Financial Stud.* 32(7):2720–2774.
- Ashraf R, Jayaraman N, Ryan HE (2012) Do pension-related business ties influence mutual fund proxy voting? Evidence from shareholder proposals on executive compensation. *J. Financial Quant. Anal.* 47(3):567–588.
- Bach L, Metzger D (2019) How close are close shareholder votes? *Rev. Financial Stud.* 32(8):3183–3214.
- Baik B, Kang J-K, Kim J-M (2010) Local institutional investors, information asymmetries, and equity returns. *J. Financial Econ.* 97(1):81–106.
- Becker B, Ivković Z, Weisbenner S (2011) Local dividend clienteles. *J. Finance* 66(2):655–683.
- Bernheim BD (1994) A theory of conformity. *J. Political Econ.* 102(5):841–877.
- Bloomberg (2016) Vanguard, BlackRock seen seldom challenging CEO pay plans. (February 17). Accessed April 20, 2020, <https://www.bloomberg.com/news/articles/2016-02-17/vanguard-blackrock-seen-seldom-challenging-companies-on-ceo-pay?sref=2CZ87tj2>.
- Bloomberg (2019) BlackRock, Vanguard face shareholder rebuke over climate votes. (December 13). Accessed April 20, 2020, <https://www.bloomberg.com/news/articles/2019-12-13/blackrock-vanguard-face-shareholder-rebuke-over-climate-votes?sref=2CZ87tj2>.
- Brickley JA, Lease RC, Smith CW Jr (1988) Ownership structure and voting on antitakeover amendments. *J. Financial Econ.* 20(1–2):267–291.
- Brown J, Ivković Z, Smith P, Weisbenner S (2008) Neighbors matter: Causal community effects and stock market participation. *J. Finance* 63(3):1509–1531.
- Butler AW, Gurun U (2012) Educational networks, mutual fund voting patterns, and CEO compensation. *Rev. Financial Stud.* 25(8):2533–2562.
- Calluzzo P, Kedia S (2019) Mutual fund board connections and proxy voting. *J. Financial Econ.* 134(3):669–688.
- Cameron AC, Gelbach JB, Miller DL (2008) Bootstrap-based improvements for inference with clustered errors. *Rev. Econom. Statist.* 90(3):414–427.
- Cao J, Liang H, Zhan X (2019) Peer effects of corporate social responsibility. *Management Sci.* 65(12):5487–5503.
- Chen X, Harford J, Li K (2007) Monitoring: Which institutions matter? *J. Financial Econ.* 86(2):279–306.
- Coval JD, Moskowitz TJ (1999) Home bias at home: Local equity preference in domestic portfolios. *J. Finance* 54(6):2045–2073.
- Coval JD, Moskowitz TJ (2001) The geography of investment: Informed trading and asset prices. *J. Political Econ.* 109(4):811–841.
- Cuñat V, Gine M, Guadalupe M (2012) The vote is cast: The effect of corporate governance on shareholder value. *J. Finance* 67(5):1943–1977.
- Cvijanović D, Dasgupta A, Zachariadis K (2016) Ties that bind: How business connections affect mutual fund activism. *J. Finance* 71(6):2933–2966.
- Daniel K, Grinblatt M, Titman S, Wermers R (1997) Measuring mutual fund performance with characteristic-based benchmarks. *J. Finance* 52:1035–1058.
- Davis GF, Kim EH (2007) Business ties and proxy voting by mutual funds. *J. Financial Econ.* 85(2):552–570.
- Del Guercio D, Hawkins J (1999) The motivation and impact of pension fund activism. *J. Financial Econ.* 52(3):293–340.
- Dimmock SG, Gerken WC, Graham NP (2018a) Is fraud contagious? Co-worker influence on misconduct by financial advisors. *J. Finance* 73(1):1417–1450.
- Dimmock SG, Gerken WC, Ivković Z, Weisbenner S (2018b) Capital gains lock-in and governance choices. *J. Financial Econ.* 127(3):113–135.
- Dougal C, Parsons CA, Titman S (2015) Urban vibrancy and corporate growth. *J. Finance* 70(1):163–210.
- Financial Times (2016) BlackRock slammed over too many votes for high pay. (May 22).
- Fortune (2016) Overpaid CEOs are being supported by mutual funds. (February 18).
- Gaspar J-M, Massa M, Matos P (2005) Shareholder investment horizons and the market for corporate control. *J. Financial Econ.* 76(1):153–165.
- Gillan SL, Starks LT (2000) Corporate governance proposals and shareholder activism: The role of institutional investors. *J. Financial Econ.* 57(2):275–305.
- Giroud X (2013) Proximity and investment: Evidence from plant-level data. *Quart. J. Econ.* 128(2):861–915.
- Glaeser E, Sacerdote B, Scheinkman J (2003) The social multiplier. *J. Eur. Econ. Assoc.* 1(2–3):345–353.
- Grinblatt M, Keloharju M (2001) How distance, language, and culture influence stockholdings and trades. *J. Finance* 56(3):1053–1073.
- Harford J, Jenter D, Li K (2011) Institutional cross-holdings and their effect on acquisition decisions. *J. Financial Econ.* 99(1):27–39.
- He J, Huang J, Zhao S (2019) Internalizing governance externalities: The role of institutional cross-ownership. *J. Financial Econ.* 134(2):400–418.
- Hong H, Kubik J, Stein JC (2005) Thy neighbor's portfolio: Word-of-mouth effects in the holdings and trades of money managers. *J. Finance* 60(6):2801–2824.
- Huson M, Parrino R, Starks LT (2001) Internal monitoring mechanisms and CEO turnover: A long-term perspective. *J. Finance* 56(6):2265–2297.
- Iliev P, Lowry M (2015) Are mutual funds active voters? *Rev. Financial Stud.* 28(12):446–485.
- Iliev P, Kalodimos J, Lowry M (2021) Investors' attention to corporate governance. *Rev. Financial Stud.* 34(2):5581–5628.
- Ivković Z, Weisbenner S (2005) Local does as local is: Information content of the geography of individual investors' common stock investments. *J. Finance* 60(1):267–306.
- Ivković Z, Weisbenner S (2007) Information diffusion effects in individual investors' common stock purchases: Covet thy neighbors' investment choices. *Rev. Financial Stud.* 20(4):1327–1357.
- Kling JR, Ludwig J, Katz LF (2005) Neighborhood effects on crime for female and male youth: Evidence from a randomized housing voucher experiment. *Quart. J. Econ.* 120(1):87–130.

- Lee AH (2021) Every vote counts: The importance of fund voting and disclosure. Speech at the 2021 ICI Mutual Funds and Investment Management Conference. Accessed April 20, 2020, <https://www.sec.gov/news/speech/lee-every-vote-counts>.
- Lerner J (1995) Venture capitalists and the oversight of private firms. *J. Finance* 50(1):301–318.
- Lerner J, Malmendier U (2013) With a little help from my (random) friends: Success and failure in post-business school entrepreneurship. *Rev. Financial Stud.* 26(10):2411–2452.
- Malenko N, Shen Y (2016) The role of proxy advisory firms: Evidence from a regression-discontinuity design. *Rev. Financial Stud.* 29(12):3394–3427.
- Malloy CJ (2005) The geography of equity analysis. *J. Finance* 60(2): 719–755.
- Manski CF (1993) Identification of endogenous social effects: The reflection problem. *Rev. Econom. Stud.* 60(3):531–542.
- Matvos G, Ostrovsky M (2008) Cross-ownership, returns, and voting in mergers. *J. Financial Econom.* 89(3):391–403.
- Matvos G, Ostrovsky M (2010) Heterogeneity and peer effects in mutual fund proxy voting. *J. Financial Econom.* 98(1):90–112.
- McCahery JA, Sautner Z, Starks LT (2016) Behind the scenes: The corporate governance preferences of institutional investors. *J. Finance* 71(2):2905–2932.
- McCrary J (2008) Testing for manipulation of the running variable in the regression discontinuity design. *J. Econometrics* 142(6):698–714.
- Mian A (2006) Distance constraints: The limits of foreign lending in poor economies. *J. Finance* 61(3):1465–1505.
- Morrow Sadali (2018) Institutional Investor Survey 2018. Accessed April 20, 2020, <https://morrowSadali.com/news-events/articles/institutional-investor-survey-2018>.
- New York Times* (2005) Potluck à la Wall Street: 'Idea dinners.' (May 24).
- New York Times* (2016a) BlackRock wields its big stick like a wet noodle on C.E.O. pay. (April 15).
- New York Times* (2016b) Your mutual fund has your proxy, like it or not (September 23).
- Parrino R, Sias RW, Starks LT (2003) Voting with their feet: Institutional ownership changes around forced CEO turnover. *J. Financial Econom.* 68(1):3–46.
- Parsons CA, Sulaeman J, Titman S (2018) The geography of financial misconduct. *J. Finance* 73(5):2087–2137.
- Petersen MA, Rajan RG (2002) Does distance still matter? The information revolution in small business lending. *J. Finance* 57(6): 2533–2570.
- Pool VK, Stoffman N, Yonker SE (2015) The people in your neighborhood: Social interactions and mutual fund portfolios. *J. Finance* 70(6):2679–2732.
- Rabin KH (2013) Examining the market power and impact of proxy advisory firms. Written statement to Committee on Financial Services United States House of Representatives. Accessed April 20, 2020, <https://www.glasslewis.com/examining-the-market-power-and-impact-of-proxy-advisory-firms/>.
- Reuters* (2016) Investors want mutual funds to get tougher on CEO pay: Reuters/Ipsos poll. (April 27). Accessed April 20, 2020, <https://www.reuters.com/article/us-ceopay-investors/investors-want-mutual-funds-to-get-tougher-on-ceo-pay-reuters-ipsos-poll-idUSKCN0X00A0>.
- Shue K (2013) Executive networks and firm policies: Evidence from the random assignment of MBA peers. *Rev. Financial Stud.* 26(6):1401–1442.
- Sommer AA (1992) Corporate governance: The search for solutions. *Univ. San Francisco Law Rev.* 26(4):695–716.
- Stein JC (2002) Information production and capital allocation: Decentralized vs. hierarchical firms. *J. Finance* 57(5):1891–1921.
- Sufi A (2007) Information asymmetry and financing arrangements: Evidence from syndicated loans. *J. Finance* 62(2):629–668.
- The Nation* (2016) Here's why we can't rely on shareholders to fix CEO pay. (May 4).
- Zimmerman D (2003) Peer effects in academic outcomes: Evidence from a natural experiment. *Rev. Econom. Statist.* 85(1):9–23.